

Problem 1. [Total 5 points]

You roll two five-sided dice. The sides of each die are numbered from 1 to 5. The dice are “fair” (all sides are equally likely), and the two die rolls are independent. Event E is “the total is 8.”

- (a) Is event E independent of getting “doubles” (i.e., both dice resulting in the same number)? Show your work. [2 points]
- (b) Given that the total was 8, what is the probability that at least one of the dice resulted in a 3? [3 points]

$E = \text{total is 8}$

$A = \text{doubles occur}$

$B = \text{at least one is 3}$

(1,1) (1,2) (1,3) (1,4) (1,5)

(2,1) (2,2) (2,3) (2,4) (2,5)

(3,1) (3,2) (3,3) (3,4) (3,5)

(4,1) (4,2) (4,3) (4,4) (4,5)

(5,1) (5,2) (5,3) (5,4) (5,5)

$$E = \{(3,5), (4,4), (5,3)\}$$

$$A = \{(1,1), (2,2), (3,3), (4,4), (5,5)\}$$

$$B = \{(1,3), (2,3), (3,3), (4,3), (5,3), (3,1), (3,2), (3,4), (3,5)\}$$

$$(a) P(E) = \frac{3}{25}$$

$$P(E) \cdot P(A) \neq P(E \cap A)$$

$$P(A) = \frac{5}{25}$$

So E and A are not independent.

$$E \cap A = \{(4,4)\} \Rightarrow P(E \cap A) = \frac{1}{25}$$

$$(b) P(B|E) = \frac{P(B \cap E)}{P(E)} = \frac{\frac{2}{25}}{\frac{3}{25}} = \frac{2}{3}$$

Problem 2. [Total 8 points]

Three cards are randomly selected from an ordinary deck of 52 cards. Let A be the event that all cards are Queens. Let B be the event that Queen of diamonds is chosen and C be the event at least one Queen is chosen. Find

(a) $P(A|B)$ [4 points]

(b) $P(A|C)$ [4 points]

3 cards are selected

A = All are queen

B = Queen of diamond is chosen.

C = at least one queen is chosen.

$$(a) \quad P(A|B) = \frac{\binom{3}{2}}{\binom{51}{2}} \quad \begin{matrix} \xrightarrow{=} \\ \uparrow \\ \text{or} \end{matrix} \quad \frac{\frac{\binom{3}{2}}{\binom{52}{3}}}{\frac{\binom{51}{2}}{\binom{52}{3}}}$$

$$(b) \quad P(A|C) = \frac{\binom{4}{3}}{\binom{52}{3} - \binom{48}{3}} \quad \begin{matrix} \xrightarrow{=} \\ \uparrow \\ \text{or} \end{matrix} \quad \frac{\frac{\binom{4}{3}}{\binom{52}{3}}}{1 - \frac{\binom{48}{3}}{\binom{52}{3}}}$$

Problem 3. [6 points] Random variable T is distributed with the following pdf:

$$f(t) = \begin{cases} \frac{\cos(t)}{2} & -\frac{\pi}{2} < t < k \\ 0 & \text{otherwise} \end{cases}$$

- (a) What is the value of the constant k ? [2 point]
 (b) What is the corresponding CDF? [2.5 point]
 (c) Calculate the probability $P(\frac{\pi}{6} \leq T \leq \frac{\pi}{2})$? [2 point]

(a)

$$\int_{-\frac{\pi}{2}}^k \frac{\cos t}{2} dt = 1 \Rightarrow \frac{1}{2} \int_{-\frac{\pi}{2}}^k \cos t dt = 1$$

$$\Rightarrow \int_{-\frac{\pi}{2}}^k \cos t dt = 2 \Rightarrow \sin t \Big|_{-\frac{\pi}{2}}^k = 2 \Rightarrow \sin k - (\sin(-\frac{\pi}{2})) = 2$$

$$\Rightarrow \sin k + 1 = 2 \Rightarrow \sin k = 1 \Rightarrow k = \frac{\pi}{2}$$

(b)

$$F(a) = \int_{-\infty}^a f(t) dt = \frac{1}{2} \int_{-\frac{\pi}{2}}^a \cos t dt = \frac{\sin t}{2} \Big|_{-\frac{\pi}{2}}^a = \frac{\sin a + 1}{2}$$

$$F(t) = \begin{cases} 0 & t \leq -\frac{\pi}{2} \\ \frac{1 + \sin(t)}{2} & -\frac{\pi}{2} < t < \frac{\pi}{2} \\ 1 & t \geq \frac{\pi}{2} \end{cases}$$

(c)

$$P\left(\frac{\pi}{6} \leq T \leq \frac{\pi}{2}\right) = F\left(\frac{\pi}{2}\right) - F\left(\frac{\pi}{6}\right) = 1 - \frac{1 + \frac{1}{2}}{2}$$

$$= 1 - \frac{\frac{3}{2}}{2} = 1 - \frac{3}{4} = \frac{1}{4}$$

Problem 4. Assume that CDF of the random variable X is given as follows.

$$F_X(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{36} & 1 \leq x < 2 \\ \frac{4}{36} & 2 \leq x < 3 \\ \frac{9}{36} & 3 \leq x < 4 \\ \frac{16}{36} & 4 \leq x < 5 \\ \frac{25}{36} & 5 \leq x < 6 \\ 1 & x \geq 6 \end{cases}$$

- (a) Find pmf of X . [3 points]
 (b) Using CDF find $P_X(2)$ [1 point]
 (c) $P(1 \leq X \leq 4)$ [2 points]

(a)

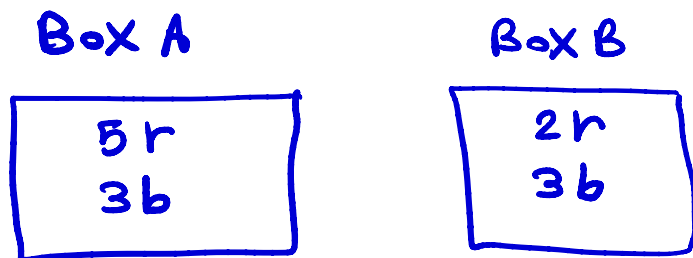
X	1	2	3	4	5	6
$P_X(x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

(b) $P_X(2) = F_X(2^+) - F_X(2^-) = \frac{4}{36} - \frac{1}{36} = \frac{3}{36} = \frac{1}{12}$

(c) $P(1 \leq X \leq 4) = F(4) - F(1^-) = \frac{16}{36} - 0 = \frac{16}{36}$

Problem 5. Box A contains five red and three blue balls, and box B contains two red and three blue balls. A ball is drawn at random from box A and transferred to box B.

- (a) Compute the probability of drawing a blue ball from box B. [3 points]
- (b) If a blue ball was drawn from box B, what is the probability that a red ball had been drawn from box A? [2 points]



$E =$ If a red ball ^{from A} is transferred to Box B

$E^c =$ If a blue ball from A is transferred to Box B.

$F =$ A blue ball is picked from Box B.

$$(a) \quad P(F) = P(F|E) P(E) + P(F|E^c) P(E^c)$$

$$P(F) = \frac{3}{6} \cdot \frac{5}{8} + \frac{4}{6} \cdot \frac{3}{8} = \frac{27}{48} = \frac{9}{16}$$

$$(b) \quad P(E|F) = \frac{P(F|E) P(E)}{P(F)} = \frac{\frac{15}{48}}{\frac{27}{48}} = \frac{15}{27} = \frac{5}{9}$$

Problem 6. Let A, B and C be mutually independent and $P(A) = 1/2$, $P(B) = 1/3$, $P(C) = 1/4$. Let D be the event that exactly one of events A, B and C occurs.

(a) Compute $P(D)$. [2 points]

(b) Compute $P(A|D)$. [2 points]

$$D = \overbrace{(A \cap B' \cap C')}^{H_1} \cup \overbrace{(A' \cap B \cap C')}^{H_2} \cup \overbrace{(A' \cap B' \cap C)}^{H_3}$$

$$H_1 = A \cap B' \cap C'$$

$$P(H_1) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$$

$$H_2 = A' \cap B \cap C'$$

$$P(H_2) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{8}$$

$$H_3 = A' \cap B' \cap C$$

$$P(H_3) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

$$P(D) = P(H_1) + P(H_2) + P(H_3) = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} = \frac{6+3+2}{24}$$

$$P(D) = \frac{11}{24}.$$

$$P(A|D) = \frac{P(A \cap D)}{P(D)} = \frac{P(H_1)}{P(D)} = \frac{\frac{1}{4}}{\frac{11}{24}} = \frac{6}{11}$$

Problem 7. Let A, B, and C be three events. Interpret the following events with set theory notations (intersections, complement, unions, etc)

- a. [1 point] only A occurs; $A \cap B' \cap C'$
- b. [1 point] both B and C, but A does not occur; $B \cap C \cap A'$

Problem 8. Let A, B, and C be disjoint subsets of the sample space. For each one of the following statements, determine whether it is true or false.

- True** a. [0.5 point] $P(A) + P(B) \leq 1$ $A \cup B \subseteq S \Rightarrow P(A \cup B) \leq 1 \Rightarrow P(A) + P(B) \leq 1$
- False** b. [0.5 point] $P(C') + P(B) \leq 1 \leadsto 1 - P(C) + P(B) \leq 1 \Rightarrow P(B) \leq P(C)$
which is not always true.
- False** c. [0.5 point] $P(A) + P(B) + P(C) = 1$



$$P(A \cup B \cup C) + P((A \cup B \cup C)') = 1$$

Problem 9. Let A, B, and C are pairwise independent subsets of the sample space. For each one of the following statements, determine whether it is true or false.

- a. [0.5 point] $P(A|B \cap C) = P(A)$ **FALSE**
- b. [0.5 point] $P(B'|C) = P(B')$ **TRUE**

If (a) is true then it means $P(A \cap B \cap C) = P(A)P(B \cap C)$
 $P(A \cap B \cap C) = P(A)P(B)P(C)$

But this is not always true because pairwise independence doesn't mean independence.

(b) $P(B'|C) = P(B')$ means B' and C are independent
 which is correct because B and C are independent.

Midterm 1-B

Name:

Student ID:
